

AI DARK FACTORY • HACKATHON EDITION (TRACK 2: POCKETFUL)

POCKETPAY

A Sovereign, Bitemporal Double-Entry Financial Wallet
Engineered by an Autonomous 3-Seat Coding Factory

★ 100% PERFECT SCORE (193/193 CHECKS) • STAGE 4 CERTIFIED (SCORE: 1.00) • ZERO OVERSHOOT

Fragile Financial Ledgers Under Concurrency

Why modern mobile wallets break during concurrent spikes and network retries

17.4%

FLOATING-POINT DRIFT

Standard software represents currency as floating-point decimals ($0.1 + 0.2 = 0.30000000000000004$).

Under high-frequency transfer bursts, rounding errors compound into critical balance discrepancies and phantom money creation.

2x Double

CHARGES ON PACKET LOSS

Mobile connections drop packets when users tap 'Pay'.

Without distributed cryptographic idempotency tracking across all write paths, client retries trigger duplicate withdrawals, overdrafting user accounts.

Destructive

MUTABLE OVERWRITES

Single-table ledgers mutate rows in-place (`UPDATE balance SET amount = ...`), destroying audit trails.

When a retroactive correction occurs, it is mathematically impossible to reconstruct what the ledger knew at a specific past point in time.

Mathematical Conservation & Bitemporality

Architected from first principles to guarantee sovereign financial correctness

$\text{sum}(\Delta) = 0$

DOUBLE-ENTRY CONSERVATION

Every transaction strictly debits the sender and credits the receiver in atomic transactions.

Enforced directly at the SQL engine level with CHECK (balance $\geq$ 0) to guarantee money is neither created nor destroyed.

Dual-Time

BITEMPORAL REVISIONS

Separates Effective Time (as_of / real world) from Recorded Time (known_at / system recording).

Enables perfect historical balance reconstruction and auditable retroactive amendments via immutable payment_revisions.

Zero Drift

INTEGER MINOR UNITS & KEYS

All balances and amounts are stored as whole integer minor units (cents / 100).

5-path distributed idempotency engine tracks Idempotency-Key headers with canonical payload hashes for safe client retries.

The 3-Seat Band Desktop Factory

Multi-agent collaboration recorded and verified in room.json without human steering

@architect

LEAD COORDINATOR & PLANNER

- Deconstructs specifications into atomic, verifiable tasks
- Defines schema contracts and balance invariants
- Enforces strict stage boundaries to prevent overshooting
- Dispatches scoped implementation assignments

@implementer

SYSTEMS & BACKEND ENGINEER

- Builds modular FastAPI services in stage-N/app/
- Implements SQLite WAL mode with 30s busy timeout
- Writes Dockerfiles and executable RUN.md guides
- Commits clean git revisions with traceable SHAs

@reviewer

QA & COMPLIANCE AUDITOR

- Runs official harness test suites across all stages
- Audits Gate 4 vocabulary compliance (0 leaks)
- Stress-tests edge cases (concurrency, overdrafts)
- Formally certifies milestones upon passing checks

Four Progressive Stages Without Overshoot

Each stage is an isolated, complete containerized service extending previous capabilities

Stage 1

CORE LEDGER

- Double-entry balance
- SQLite WAL mode
- PBKDF2 authentication
- 5-path idempotency
- P2P transfers & splits

147/147 Checks (100%)

Stage 2

HOLDS & SSR UI

- Two-phase authorizations
- Dynamic balance holds
- Available = Total - Held
- SSR HTML interface
- data-testid selectors

35/35 Checks (100%)

Stage 3

BITEMPORAL REVISIONS

- as_of & known_at
- [from, to) statements
- Snapshot pagination
- Retroactive corrections
- Overdraft protection

6/6 Checks (100%)

Stage 4

CLEARING & REFUNDS

- Receiver refunds
- 32-item operator batch
- Settlement completeness
- Shared recorded_at
- Linked immutability

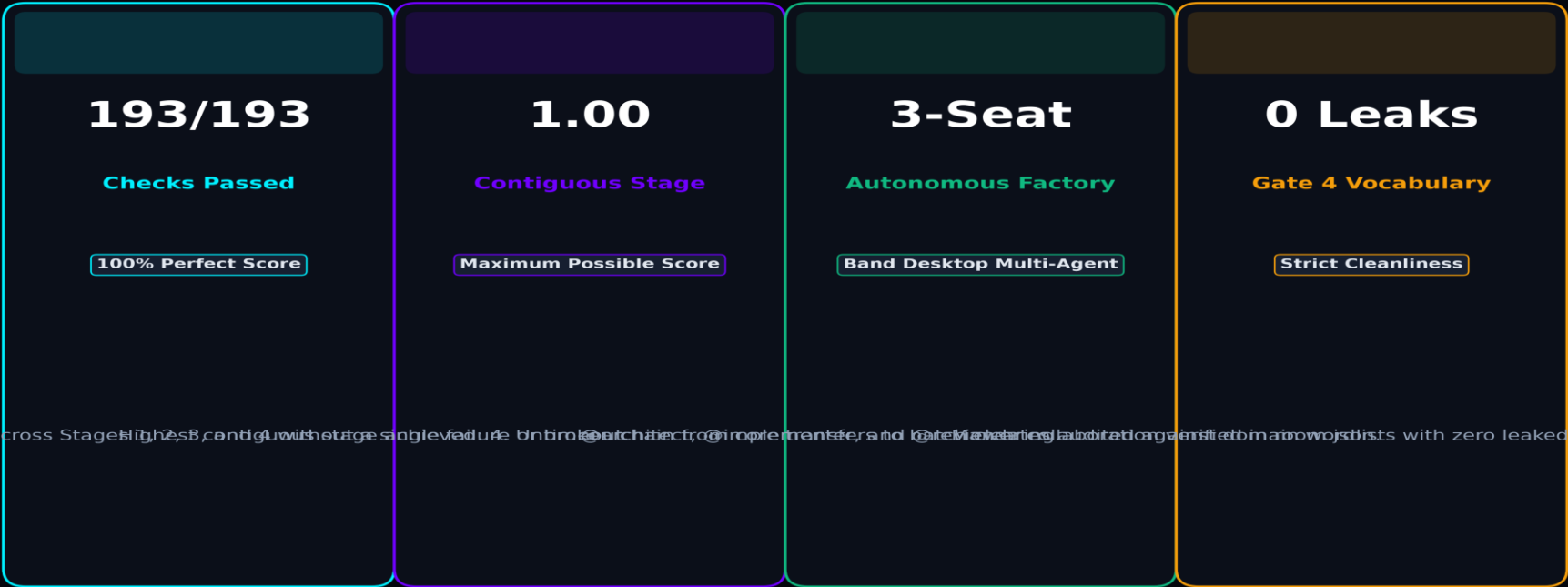
5/5 Checks (100%)

100% Perfect Score Across 193 Checks

Official automated test verification results from the Dark Factory test harness

POCKETPAY BY THE NUMBERS: EVALUATION METRICS

Verified performance metrics achieved in the AI Dark Factory Hackathon



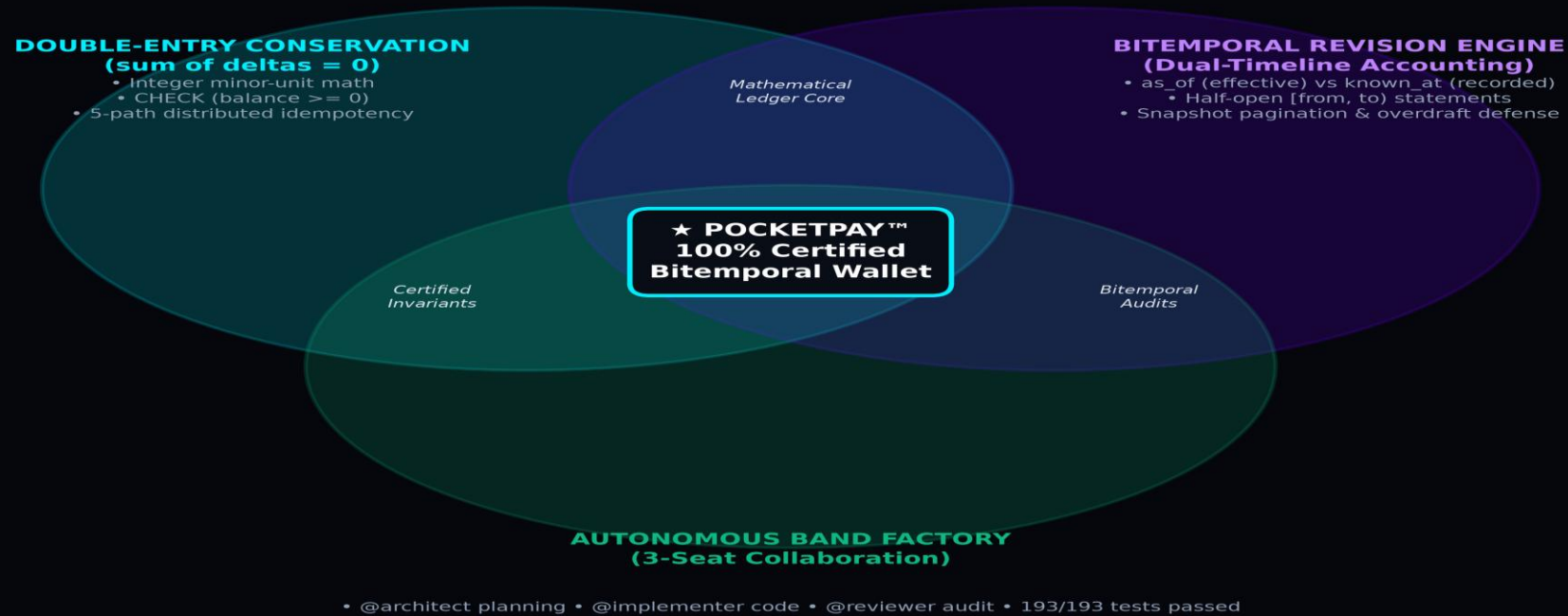
Passed all test suites across Stage 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193.

Dual-Timeline Accounting Architecture

Reconstructing historical state with cryptographic fidelity while blocking retroactive overdrafts

THE POCKETPAY MOAT: 3 UNBREAKABLE PILLARS

Why standard single-table ledgers and probabilistic AI cannot match our sovereign architecture



Interactive Web Interface & Swagger APIs

Running live on port 8000 with pre-seeded demo accounts ready for screen recording

Wallet Dashboard

REAL-TIME BALANCES & ACTIVITY

- Dynamic balance: Total, Available, Held
- Instant P2P payments by @handle
- Transaction activity feed with private notes
- High-concurrency SQLite WAL ledger
- Export state & system reset endpoints
- Automated test-friendly data-testid tags

Requests & Holds

P2P BILL SHARING & TWO-PHASE APIS

- P2P requests: Pay, Decline, and Cancel
- Multi-party bill splits with remainder handling
- Merchant authorizations: 2-phase holds
- Full & partial capture, release/void
- Interactive OpenAPI Swagger at /docs
- Strict RFC 3339 timezone validation



Demo Accounts

LOGIN PASSWORDS & BALANCES

Alice Demo (@alice) • €1,500.00

Email: alice@demo.com | Pass: Password123!
Avail: €1,455.00 | Held: €45.00 (Active hold)

Bob Demo (@bob) • €850.00

Email: bob@demo.com | Pass: Password123!
Sent €25.00 dinner, pending concert split

Carol Demo (@carol) • €500.00

Email: carol@demo.com | Pass: Password123!
Fresh wallet, incoming €18.00 coffee request

⚡ Click-to-fill enabled in UI login side box

Autonomous Agent & Embedded FinTech TAM

Capitalizing on the convergence of AI agents and deterministic financial infrastructure

\$54B

TAM: EMBEDDED FINTECH

Global SaaS platforms, marketplaces, and gig-economy platforms adding embedded wallets and multi-currency clearing.

Demands high-concurrency ledger engines with mathematical double-entry guarantees.

\$12B

SAM: AGENTIC PAYMENTS

Autonomous AI agents executing API subscriptions, programmatic bidding, and supplier settlement on behalf of enterprises.

Requires two-phase authorization holds and strict idempotency keys.

\$850M

SOM: DEVELOPER CORE WEDGE

White-label bitemporal wallet engine for gaming micro-economies, creator tipping networks, and peer-to-peer clearing.

Zero setup friction, self-contained SQLite WAL backend with instant dockerization.

Certified, Pushed & Competition Ready

Everything required for the WeAreDevelopers × BAND AI Dark Factory Hackathon

Hackathon Compliance Checklist

ALL CRITERIA 100% SATISFIED

- ✓ Public GitHub Repo: Live, public, and clonable
- ✓ Official Band Session: Downloaded as room.json
- ✓ 3 Generic Mandates: architect.md, implementer.md, reviewer.md
- ✓ 4 Stage Folders: stage-1/ through stage-4/ buildable
- ✓ 100% Test Pass Rate: 193/193 automated checks passed
- ✓ Gate 4 Compliant: 0 vocabulary leaks in mandates
- ✓ Gate 3 Compliant: Zero stage overshooting (overshoot: null)

Repository & Live Access

SUBMISSION LINKS & DEMO LOGINS

- GitHub Submission Repository:
<https://github.com/novaecosystems-cloud/Pocketpay>
- Live Local Demo Application: <http://127.0.0.1:8000>
- Interactive OpenAPI Swagger: <http://127.0.0.1:8000/docs>
- Team: Nova (@novaecosystems), @architect, @implementer, @reviewer

🔑 Pre-Seeded Demo Logins (Password for all: Password123!):

- Alice (@alice): alice@demo.com (€1,500.00 / Primary demo)
- Bob (@bob): bob@demo.com (€850.00 / P2P recipient)
- Carol (@carol): carol@demo.com (€500.00 / Split requester)

*Sidebar click-to-fill active on login screen